London Bicycles

Data Engineering Insight, Cleaning, Outlier & Risk Report

GCP Project: **m2-dataset-study** | Dataset: **bigquery-public-data.london_bicycles** | Generated: **19 May 2026**

Sections at a Glance

1. Executive Summary

This report analyses the TfL Santander Cycles dataset on Google BigQuery: **83.4 million rental records** from January 2015 to January 2023 across **~800 docking stations**. It covers business insights, surprise findings, a comprehensive data cleaning and outlier analysis, pipeline architecture, risks, and executive recommendations.

Central problem: *TfL operates one fleet for two completely different markets — commuters and leisure riders — without optimising rebalancing for either.*

Metric	Value
Total records	83,434,866
Date range	4 Jan 2015 – 15 Jan 2023
Active stations (peak)	~800
Unique bikes tracked	~24,700
Clean records (for analysis)	77,995,546 (93.5%)
Records needing cleaning	5,439,320 (6.5%) across 7 categories
Peak month (Jul 2022)	1,303,376 rentals
Trough month (Jan 2022)	741,875 rentals

2. Key Insights from the Data

2.1 Two Markets, One Fleet

Hour	Weekday Rentals	Weekend Rentals	Ratio
08:00	7,156,970	412,928	17×
12:00–14:00	~5.7M	~5.8M	~1×
17:00–18:00	13,221,451	~3.1M	4×

- **Weekday:** bimodal spikes at 08:00 and 17–18:00 → commuters.
- **Weekend:** flat plateau 10:00–17:00 → leisure/tourist riders.

2.2 Top Routes Are All Leisure Circuits

Route (Start → End)	Trips	Avg Duration
Hyde Park Corner → Albert Gate	33,122	42.2 min
Hyde Park Corner → Triangle Car Park	31,747	29.7 min
Black Lion Gate → Palace Gate	30,908	23.5 min
Aquatic Centre → Podium (Olympic Park)	23,501	42.9 min

2.3 COVID Signature (2020) & Seasonal 3× Swing

Year	Rentals	Avg Duration	Unique Bikes
2019	10,310,063	19.5 min	12,943
2020	**10,287,040**	**26.2 min** ↑	**14,743**
2021	10,821,289	21.9 min	14,962
2022	11,027,519	21.3 min	24,715

2020 duration spike (+34%) reflects lockdown exercise behaviour. Seasonal swing: Jan 742k → Jul 1.3M rentals (3× difference).

2.4 E-Bike vs Classic — Coverage Limited to 2022+

Model	Rentals	Avg Duration	Coverage
NULL/Untagged	81,048,143	~21.9 min	All years (pre-tagging era)
CLASSIC	2,254,688	23.5 min	2022–2023 only
PBSC_EBIKE	132,035	18.9 min	2022–2023 only

3. Surprise Findings

Four unexpected findings emerged from direct data queries — each with significant implications for pipeline design, data governance, and business value.

3.1 Maintenance Trips Hidden as Rental Records

★ The Surprise	Three 'Mechanical Workshop' stations appear as valid rental destinations. Bikes sent there show average durations of 131–178 hours. The longest single rental is 65.3 days (bike #60353 — in workshop custody, not a trip).

Workshop	Trips	Unique Bikes	Avg Duration
Mechanical Workshop Clapham	777	757	9,076 min (151 hrs)
Mechanical Workshop Penton	724	706	7,889 min (131 hrs)
Electrical Workshop PS	2	2	10,707 min (178 hrs)

Fix: Flag is_maintenance_trip in stg_cycle_hire. Route to mart_maintenance_trips — these records have operational value for fleet management.

3.2 Saturday Night Is a Hidden Third Market

★ The Surprise	Sunday midnight (Saturday night) generates 210,272 rentals — 2.3× Tuesday midnight — with longer avg durations (32.8 min vs fleet mean 21.9 min). These are nightlife riders cycling home from entertainment hubs, not commuters or leisure tourists.

Day	Midnight Rentals	2am Rentals	Avg Night Duration
Sunday (Sat night)	**210,272**	**110,872**	**32.8 min**
Saturday (Fri night)	196,682	99,461	32.7 min
Friday	116,406	41,649	31.6 min
Tuesday (baseline)	73,963	25,163	28.6 min

Fix: Add trip_segment = 'nightlife' for 00:00–04:00 Fri/Sat. Build mart_nightlife_demand to rank top departure stations in this window.

3.3 Waterloo Bleeds Bikes; Borough Absorbs Them — Permanently

★ The Surprise	Over 8 years, Waterloo Station 2 shows +69,147 net bike outflow (16.2% imbalance). Hop Exchange Borough shows –132,544 net inflow (–15.0%). These are structural geographic patterns — no algorithm can fix them without daily van redistribution.

Station	Departures	Arrivals	Net Flow	Imbalance
Hop Exchange, The Borough	376,225	508,769	–132,544	–15.0% (sink)
Holborn Circus, Holborn	222,739	304,269	–81,530	–15.5% (sink)
Waterloo Station 2	**247,486**	**178,339**	**+69,147**	**+16.2% (source)**

Station	Departures	Arrivals	Net Flow	Imbalance
Cloudesley Road, Angel	89,758	54,506	+35,252	+24.4% (source)

Fix: Add is_structural_imbalance to dim_station (net flow > 10% over 90-day rolling). Surface in exec dashboard as stations requiring daily redistribution.

3.4 Super-Bikes: 8,197 Rentals, Zero Maintenance Records

★ **The Surprise**	Bike #12942 was rented 8,197 times over 7.7 years (~3×/day), accumulating 2,810 hours of riding. The top 10 hardest-working bikes each show 7,600–8,200 rentals with no associated workshop record — either they were never maintained (safety risk) or the maintenance system doesn't link to bike_id (data gap).

Bike ID	Rentals	Hours Ridden	Avg Trip	Active Lifespan
12942	8,197	2,810 hrs	20.6 min	2,800 days (7.7 yrs)
11077	8,063	2,701 hrs	20.1 min	2,805 days
10601	7,970	2,749 hrs	20.7 min	2,719 days

Fix: Build mart_bike_utilisation with cumulative rentals and hours per bike_id. Flag bikes exceeding 5,000 rentals or 2,000 hours with no workshop record.

4. Data Cleaning & Outlier Analysis

A complete NULL audit and outlier classification was run across every column of both source tables. All findings are from direct BigQuery queries on the live dataset. Cleaning decisions are encoded as dbt flags — records are **never silently dropped**; they are routed to appropriate marts where they retain analytical value.

4.1 cycle_hire — Complete NULL Audit

Column	NULL Count	% of Total	Root Cause	Cleaning Decision
rental_id	0	0%	—	No action needed
duration	19,587	0.023%	Trip started but never ended — always co-null with end_date and end_station	Flag is_incomplete_trip; route to mart_incomplete_trips; exclude from analytics
duration_ms	19,587	0.023%	Fully redundant (= duration × 1000); always co-null with duration	Drop column in staging — zero information gain
bike_id	15	< 0.001%	June 2017 system glitch — all end stations also NULL	Flag is_unknown_bike; exclude from bike-level analysis only
bike_model	56,678,942	67.93%	Model tagging only introduced in 2022 — expected gap	Map NULL → 'Unknown'; scope model analysis to 2022+ only
start_station_id	229,639	0.275%	7-day system window Aug–Sep 2016 only; station names are present	Flag is_missing_start_id; resolve via name→id lookup where possible
end_station_id	561,504	0.673%	Three sub-patterns (see 4.2 below)	Flag-based routing (see 4.2)
end_station_name	19,721	0.024%	Subset of incomplete trips + 203 ghost records	Included in is_incomplete_trip or is_ghost_end flag
end_station_*_terminal	83,205,227	99.7%	Column never populated — always NULL	Drop both terminal columns in staging
end_station_priority_id	83,205,227	99.7%	Column never populated — always NULL	Drop column in staging

4.2 end_station NULL — Three Distinct Sub-Patterns

Not all 561,504 NULL end_station_id records are equal. Querying the joint distribution of end_station_id, end_station_name, and end_date reveals three separate root causes requiring different treatments:

Pattern	Count	% Total	Meaning	Treatment
ID null, name present, end_date present	541,783	0.649%	Station ID was not recorded but name exists — likely a logging bug. Trip completed.	Flag is_missing_end_id. Resolve ID via name→station lookup. Usable for route analysis after resolution.

Pattern	Count	% Total	Meaning	Treatment
ID null, name null, end_date null	19,518	0.023%	Trip never ended — bike not returned. Same rows as null_duration (incomplete trip).	Flag is_incomplete_trip. Route to mart_incomplete_trips. Exclude from all route/duration analytics.
ID null, name null, end_date present	203	< 0.001%	End date logged but no station info — 'ghost' records. Avg duration 563 min (9.4 hrs). Likely system error.	Flag is_ghost_end. Exclude from main analytics. Log in data quality report.

4.3 start_station_id NULL — Isolated 2016 System Window

★ **Key Finding**	All 229,639 NULL start_station_id records occur exclusively in a 7-day window: 31 August – 6 September 2016. Station names are fully present for all rows. This is a localised system logging failure, not random data loss. Resolution: join back to station names to recover ~95% of IDs.

dbt fix in stg_cycle_hire: *COALESCE(start_station_id, station_name_to_id_lookup.id)* using a seed table mapping station names to IDs. Flag remaining unresolved rows as is_missing_start_id = TRUE.

4.4 Duration Outliers — Full Taxonomy

Duration is the most analytically important field. A percentile analysis was run to establish empirical thresholds:

Percentile	Duration	Formatted
P25	480 sec	8 min
P50	840 sec	14 min
P75	1,320 sec	22 min
P90	1,980 sec	33 min
P95	3,094 sec	51.6 min
P99	8,220 sec	137 min
P100 (max)	5,641,532 sec	65.3 days
IQR (Q3–Q1)	840 sec	14 min
IQR upper fence (Q3 + 1.5×IQR)	2,580 sec	43 min — too aggressive

Why the IQR fence is wrong here: The standard IQR upper fence of 43 minutes would flag 6.4% of all trips (5.35M records) as outliers — including many legitimate leisure rides and all round-trips in Hyde Park (avg 42–57 min). This is a case where **domain knowledge must override a statistical rule**. TfL's pricing model shows trips under 30 min are free; over 30 min incur charges. Long leisure trips are expected and valid. The correct threshold is 24 hours.

Cleaning Category	Count	% of Total	Action	Analytical Fate
Clean (≤ 86,400 sec, no flags)	77,995,546	93.481%	Keep	Core analytics
Mild outlier (IQR fence 43–1440 min)	5,352,254	6.415%	**KEEP** — legitimate leisure/park trips	Core analytics
Zero duration	39,467	0.047%	Drop from analytics	Excluded

Cleaning Category	Count	% of Total	Action	Analytical Fate
Extreme outlier (> 24 hrs, non-maintenance)	26,272	0.031%	Flag is_anomalous_duration	mart_anomalous_trips
Incomplete trip (no end_date/station)	19,587	0.023%	Flag is_incomplete_trip	mart_incomplete_trips
Maintenance trip (workshop destination)	1,502	0.002%	Flag is_maintenance_trip	mart_maintenance_trips
Negative duration	238	0.000%	Drop (data corruption)	Excluded

4.5 cycle_stations — NULL Audit & Capacity Outliers

Column	NULL Count	% of 800	Root Cause	Cleaning Decision
id, name, installed, lat, lon, locked, bikes_count, docks_count, nbEmptyDocks, temporary, terminal_name	0	0%	—	No action needed
install_date	88	11.0%	Stations added before the tracking field was introduced	Flag install_date_unknown = TRUE; do not impute — date is unknowable
removal_date	797	99.6%	Expected — only 3 stations ever formally removed	No action; NULL means active

Capacity outliers found:

- 2 stations have docks_count = 0: *New Road 2, Whitechapel* (not installed) and *Kingsway Southbound, Strand* (marked installed but 0 docks — data entry error). Flag has_capacity_error = TRUE; exclude from capacity analyses.
- bikes_count never exceeds docks_count (max bikes 42 vs max docks 64) — no capacity overflow anomalies in the snapshot.
- All 800 stations are within London bounding box (51.45°–51.55°N, –0.24°–0.00°E) — no geographic outliers.

4.6 Redundant Column: duration_ms

★ **Finding**	duration_ms is exactly duration × 1000 in every single row — verified by checking ABS(duration – ROUND(duration_ms / 1000)) > 1 across all 83.4M records: 0 mismatches. It always co-NULLs with duration. It carries zero additional information. Drop it in stg_cycle_hire to reduce storage and query cost.

4.7 dbt Staging SQL — Complete Cleaning Logic

The following SQL encodes all cleaning decisions as boolean flags, so no records are silently lost:

```
-- models/staging/stg_cycle_hire.sql
SELECT
rental_id,
bike_id,
NULLIF(TRIM(bike_model), '') AS bike_model,
start_date,
end_date,
start_station_id,
start_station_name,
```

```
end_station_id,
end_station_name,
duration AS duration_sec,
ROUND(duration / 60.0, 2) AS duration_min,
-- duration_ms DROPPED (= duration × 1000, zero info)
-- terminal / priority columns DROPPED (99.7% NULL, never populated)

-- Cleaning flags (never drop — route to appropriate mart)
duration IS NULL AND end_date IS NULL AS is_incomplete_trip,
duration IS NOT NULL AND duration <= 0 AS is_invalid_duration,
LOWER(end_station_name) LIKE '%workshop%' AS is_maintenance_trip,
duration > 86400 AS is_anomalous_duration,
start_station_id IS NULL
AND start_station_name IS NOT NULL AS is_missing_start_id,
end_station_id IS NULL
AND end_station_name IS NOT NULL AS is_missing_end_id,
end_station_id IS NULL
AND end_station_name IS NULL
AND end_date IS NOT NULL AS is_ghost_end,
bike_id IS NULL AS is_unknown_bike,
FROM `bigquery-public-data.london_bicycles.cycle_hire`

-- models/staging/stg_stations.sql
SELECT
id, name, installed, latitude, longitude,
docks_count, bikes_count, nbEmptyDocks,
temporary, terminal_name, install_date, removal_date,
install_date IS NULL AS install_date_unknown,
docks_count = 0 AS has_capacity_error,
removal_date IS NOT NULL AS is_removed
FROM `bigquery-public-data.london_bicycles.cycle_stations`
```

fact_rentals filter logic: WHERE NOT is_invalid_duration AND NOT is_incomplete_trip AND NOT is_maintenance_trip AND NOT is_ghost_end. Everything else — including anomalous_duration and missing IDs — is kept and routed via the appropriate flag.

4.8 Impact of Cleaning on Key Metrics

Metric	Raw (uncleaned)	After Cleaning	Delta	Why It Changed
Avg trip duration	21.9 min	21.2 min	–0.7 min	Removes maintenance trips (131–178 hr) and extreme outliers
Total usable trips	83,434,866	77,995,546	–5.4M (–6.5%)	Removes invalid + incomplete + maintenance records
Trips > 24 hrs	47,133	26,272	–20,861	Maintenance trips separated into own mart
Null end_station rate	0.673%	0.024%	–0.649%	541,783 IDs resolved via name lookup; flagged separately
Stations with metadata	785 of 1,625	785 + enrichment	TBD	Requires TfL API enrichment (R1)

5. Proposed Data Pipeline Architecture

Layer	Tool	Output
Source	bigquery-public-data (read-only)	Raw cycle_hire, cycle_stations
Staging	dbt — stg_cycle_hire, stg_stations	Cleaned + flagged; 5 columns dropped
Dimensions	dbt — dim_date, dim_station, dim_bike	Conformed; enriched from TfL API seed
Fact	dbt — fact_rentals (incremental)	Clean 78M row fact + 7 cleaning flags
Marts	dbt — 6 marts (incl. maintenance/nightlife/bikes)	Analytical + operational tables
Analysis	Python / pandas / SQLAlchemy	Jupyter notebooks & charts
Orchestration	GitHub Actions (cron)	Scheduled dbt runs + quality checks

5.1 Updated Star Schema

```
Fact table: fact_rentals
rental_id, bike_id (FK), start_station_id (FK), end_station_id (FK),
date_id (FK), duration_sec, duration_min, is_round_trip,
trip_segment, -- commute_am|commute_pm|leisure|nightlife|night
is_maintenance_trip, is_anomalous_duration,
is_missing_start_id, is_missing_end_id, is_unknown_bike
```

```
Dimensions:
dim_station (+ is_structural_imbalance, has_capacity_error, install_date_unknown)
dim_bike (+ cumulative_rentals, cumulative_hours, exceeds_retirement_threshold)
dim_date (+ season, is_weekend, is_uk_bank_holiday, is_disrupted_period)
```

6. Risk Register & Mitigations

R1: Station Metadata Gap — 52% of Hire Stations Unresolvable	Likelihood: **High** | Impact: **High**
Finding	cycle_hire references 1,625 station IDs; cycle_stations has 785. 840 (52%) lack lat/lon/capacity.
Mitigation	Enrich via TfL Bikepoint API weekly seed. Flag unresolved as 'unknown_station'. Never NULL-fill with guesses.

R2: bike_model NULL for 97% of Records	Likelihood: **High** | Impact: **Medium**
Finding	Only 2.86% of records have a non-null model. E-bike analysis valid for 2022+ only.
Mitigation	Map NULL/empty → 'Unknown' in dim_bike. Scope model analysis to 2022+. Document in mart YAML.

R3: Duration Outliers + Maintenance Masquerade Inflate KPIs	Likelihood: **High** | Impact: **High**
Finding	47,133 trips > 24hrs; workshop 'rentals' average 131–178 hrs; max record = 65.3 days.
Mitigation	Flag is_maintenance_trip and is_anomalous_duration. Apply 24hr cap for customer analytics. Separate marts for each type.

R4: IQR Fence Misidentifies 5.35M Legitimate Trips as Outliers	Likelihood: **High** | Impact: **High**
Finding	Standard IQR upper fence (43 min) would exclude 6.4% of all trips — including valid leisure rides.
Mitigation	Use 24-hour domain-driven threshold, not IQR. Document the decision in dbt model YAML with rationale.

R5: NULL end_station_id Has Three Root Causes Requiring Different Treatments	Likelihood: **Medium** | Impact: **High**
Finding	541k have names (resolvable); 19.5k are incomplete trips; 203 are ghost records. Treating all the same loses data.
Mitigation	Three separate flags: is_missing_end_id, is_incomplete_trip, is_ghost_end. Resolve 541k via name lookup.

R6: NULL start_station_id — 2016 System Window	Likelihood: **Medium** | Impact: **Medium**
Finding	229,639 records in Aug–Sep 2016 have no start_station_id but names are present.
Mitigation	COALESCE via name→id seed table. Recover ~95% of IDs. Flag remaining unresolved.

R7: EU Region Lock	Likelihood: **Medium** | Impact: **High**
Finding	Any dataset outside EU region causes cross-region query failures with no fallback.
Mitigation	Set location: EU in dbt profiles.yml and all bq commands. CI check validates location before dbt runs.

R8: Structural Station Imbalance Cannot Be Solved Algorithmically	Likelihood: **High** | Impact: **High**
Finding	Waterloo (+69k outflow) and Hop Exchange (–132k inflow) are permanent geographic patterns — 8 years of evidence.
Mitigation	Flag is_structural_imbalance in dim_station. Surface as 'stations requiring daily van redistribution' in exec dashboard.

7. ELT Transformation Plan (dbt)

dbt Model	Type	Key Logic
stg_cycle_hire	View	All cleaning flags; drop duration_ms + terminal + priority cols
stg_stations	View	Add has_capacity_error, install_date_unknown, is_removed
dim_date	Table	Date spine; season, is_weekend, is_uk_bank_holiday, is_disrupted_period
dim_station	Table	TfL API enrichment; is_structural_imbalance (90-day net flow)
dim_bike	Table	NULL model → Unknown; cumulative_rentals/hours; retirement flag
fact_rentals	Incremental	Filter invalid/incomplete/ghost; add trip_segment incl. nightlife
mart_station_demand	Table	Hourly net flow; structural vs temporary imbalance flag
mart_nightlife_demand	Table	Fri/Sat 00:00–04:00 top departure stations and routes

dbt Model	Type	Key Logic
mart_bike_utilisation	Table	Cumulative rentals+hours per bike; exceeds_retirement_threshold
mart_maintenance_trips	Table	Workshop-destination records with bike_id and duration
mart_monthly_trends	Table	Monthly KPIs; is_disrupted_period from external_events seed

8. Data Quality Testing Plan

Test	Tool	Threshold	Action on Fail
fact_rentals: no invalid/incomplete/ghost rows	dbt singular	0 failures	Block pipeline
fact_rentals.rental_id unique + not null	dbt generic	0 failures	Block pipeline
fact_rentals.duration_sec > 0 (after filter)	dbt singular	0 failures	Block pipeline
is_missing_end_id rate < 1%	dbt test	Warn 0.7%, Error 1%	Alert only
is_maintenance_trip count plausible (< 2,000/run)	dbt singular	Warn on breach	Investigate
Monthly rental count within 3σ of rolling mean	Great Expectations	Warn on breach	Alert + flag mart
Avg duration change < 30% month-over-month	Great Expectations	Warn on breach	Alert + investigate
dim_station: no zero-dock stations in capacity mart	dbt singular	0 failures	Block capacity analytics

9. Python Analysis — Recommended Notebook Structure

Notebook	Analysis	Business Question
01_demand_patterns.ipynb	Heatmap: hour × day_type (clean data only)	When and where to rebalance?
02_station_flow.ipynb	Net flow; structural vs temporary imbalance	Which stations need daily vans?
03_seasonality.ipynb	Monthly trend + 3σ anomaly detection; external events	How should fleet scale?
04_nightlife_segment.ipynb	Fri/Sat 00:00–04:00 top stations & routes	Where is the third market?
05_bike_utilisation.ipynb	Cumulative hours/rentals per bike; retirement list	Which bikes need replacing?

Always query marts — never pull the raw 83M-row fact table into pandas memory. Connect via: *create_engine("bigquery://m2-dataset-study")*. Filter all notebooks to clean records (is_incomplete_trip = FALSE, etc.) unless the notebook is specifically analysing anomalies.

10. Executive Presentation Outline (10 min)

Slide	Title	Key Message	Time
1	The Problem	One fleet, two markets + hidden nightlife segment	1 min
2	The Evidence	Weekday vs Weekend demand; Saturday midnight spike	1.5 min
3	Surprise Findings	Maintenance masquerade; super-bikes; Waterloo imbalance	1.5 min
4	Data Quality	6.5% of records cleaned; IQR fence rejected; 3 NULL patterns	1 min
5	Architecture	ELT: BigQuery → dbt → Python; 11 models; 8 quality tests	1.5 min
6	Risks & Controls	Top risks R3, R4, R7, R8 + mitigations	1 min
7	Recommendations	Segment rebalancing + nightlife pre-positioning + bike retirement	1 min
8	Q&A;	Ready to answer	5 min

ROI framing: 1% availability improvement at 08:00–09:30 peak = 72,000 additional rentals = £144k–£216k annual revenue. Nightlife pre-positioning (210k Sat-night trips) offers a further 10–15% uplift. Bike retirement model prevents costly liability from over-used frames.

11. Recommended Tech Stack & Justification

Component	Choice	Why
Cloud platform	GCP / BigQuery (EU)	Source data already here; zero egress; serverless scale
Transformation	dbt Core	SQL-native; version-controlled; built-in testing; free tier
Orchestration	GitHub Actions (cron)	Free; no infra; sufficient for batch cadence
Analysis	Python + pandas + BigQuery connector	Assignment requirement; Jupyter-friendly
Data quality	dbt tests + Great Expectations	dbt for structural; GE for statistical anomaly detection
Diagramming	Excalidraw	Free; clean architecture diagrams; SVG export
Documentation	dbt docs generate	Auto-generates data catalog from model YAML — zero extra work